MARKING SCHEME

Study-Uni

AC4101 — Management Accounting: Performance and Decision Making

Summer 2023 · Paper 1 · Full Marking Scheme

Module Code	AC4101
Module Title	Management Accounting: Performance and Decision Making
Examination Session	Summer 2023
Paper Number	Paper 1
Duration of Paper	1.5 hours
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Question	Scenario	Marks
Question 1	Vent Ltd — Standard Costing & Variance Analysis	50
Question 2	Konrad (Joe & Tom) — NPV, ARR, Payback & ESG	50
Question 3	JXC Ltd — Overhead Allocation & Balanced Scorecard	50

Instructions to Candidates (as stated on the paper): Answer ANY TWO questions. Show all workings clearly – marks will be awarded for workings.

Note on this marking scheme: Numerical parts are marked primarily on correct method and workings, with the **own-figure rule** applied throughout – an error carried through consistently does not cause repeated mark loss. Discursive parts are marked using qualitative bands assessing depth of analysis, application to the specific scenario, and clear structure. No academic citations are expected or required in any answer in a closed-book exam.

QUESTION 1

Vent Ltd

Total: 50 marks

This question tests standard costing and variance analysis for a wind turbine component manufacturer. It covers sales margin volume variance, material usage, labour efficiency, variable and fixed overhead variances, plus discursive understanding of standard vs budgetary control systems and non-financial objectives.

Parts in this Question

Part	Requirement	Marks
(a)	Calculate five variances: sales margin volume, material usage, labour efficiency, total variable OH, total fixed OH	25
(b)	TWO advantages and TWO disadvantages of replacing metal material with plastic alternative	8
(c)	Explain why budgetary control is preferred in certain environments, with example	10
(d)	THREE non-financial objectives management could set, identifying stakeholder group for each	7

Question 1(a)

25 marks

Calculate the following for the month: Sales Margin Volume Variance; Material Usage Variance; Labour Efficiency Variance; Total Variable Overhead Variance; Total Fixed Overhead Variance.

Full Model Answer

Standard cost card (Part S) and key derivations

Standard cost element	Inputs	Per unit (€)
Materials	5m × €0.20	1.00
Labour	4 hrs × €12.00	48.00
Variable overhead	4 hrs × €1.50	6.00
Fixed overhead	4 hrs × €3.50	14.00
Standard total cost		**69.00**
Standard gross profit		3.00
Standard selling price		**72.00**

Budgeted units derivation: Vent Ltd budgeted to achieve a €30,000 profit. Budgeted profit = Budgeted units × Standard gross profit per unit. Therefore: Budgeted units = €30,000 ÷ €3.00 = **10,000 units**.

Actual material usage: The paper states direct material "purchased & used" = 38,450 metres, but also that stocks of direct material *decreased by 800 metres* during the month. A decrease in stock means more material was consumed from opening stock than was replaced by purchases. Therefore: **Actual usage = 38,450 + 800 = 39,250 metres**. (Actual production = 8,250 units; finished goods stock unchanged, confirming actual sales = actual production = 8,250 units.)

(i) Sales Margin Volume Variance

Actual sales volume	8,250 units
Budgeted sales volume	10,000 units
Difference	(1,750) units
Standard gross profit per unit	€3.00
Sales Margin Volume Variance	**€5,250 Adverse**

(ii) Material Usage Variance

Standard usage for actual production (8,250 × 5m)	41,250 metres
Actual usage	39,250 metres
Favourable difference	2,000 metres
Standard material price per metre	€0.20
Material Usage Variance	**€400 Favourable**

(iii) Labour Efficiency Variance

Standard hours for actual production (8,250 × 4 hrs)	33,000 hrs
Actual hours worked	34,000 hrs
Adverse difference	(1,000) hrs
Standard labour rate per hour	€12.00
Labour Efficiency Variance	**€12,000 Adverse**

(iv) Total Variable Overhead Variance

Standard variable OH for actual output (8,250 × 4 hrs × €1.50)	€49,500
Actual variable overhead incurred	€50,000
Total Variable Overhead Variance	**€500 Adverse**

(v) Total Fixed Overhead Variance

Fixed OH absorbed: actual production × std hrs × std rate (8,250 × 4 × €3.50)	€115,500
Actual fixed overhead incurred	€139,500
Total Fixed Overhead Variance	**€24,000 Adverse**

Sub-variance breakdown (for context): Budgeted fixed OH = 10,000 units × 4 hrs × €3.50 = €140,000. Fixed OH Expenditure Variance = €140,000 − €139,500 = €500 Favourable. Fixed OH Volume Variance = (8,250 − 10,000) × 4 × €3.50 = €24,500 Adverse. Total = €24,500 A + €500 F = €24,000 Adverse — reconciles to the total above.

Mark Allocation

Marks	Criteria
22–25	All five variances correctly calculated with correct F/A signs; budgeted units correctly derived from €30,000 target profit; actual material usage correctly adjusted for the 800m stock decrease; full workings shown throughout.
17–21	Four of five variances correct; own-figure rule applied; correct derivation of budgeted units and/or material usage adjustment; minor arithmetic errors in one variance.
11–16	Correct method demonstrated for most variances but with errors in the stock-movement adjustment for materials, or budgeted units incorrectly derived, causing cascading impact; three or more variances substantially correct.
5–10	Some correct variance calculations but fundamental errors in approach (e.g. using actual units instead of actual production for overhead variances, or ignoring the stock decrease); partial attempt.
0–4	No meaningful attempt, or fundamental misunderstanding of variance analysis.

Question 1(b)

8 marks

Advise the management team of Vent Ltd of TWO advantages and TWO disadvantages of replacing the metal material part currently used in production with the alternative plastic part.

Full Model Answer

Advantage 1: Environmental Sustainability and ESG Benefits

The plastic alternative is made from recycled material and is itself recyclable at end of life. For Vent Ltd, which operates in the wind turbine construction sector – an industry with strong environmental credentials – this could enhance its sustainability profile, improve ESG reporting, and help it meet any green procurement requirements set by its customers or the wind energy industry more broadly.

Advantage 2: Reduced Material Waste and Potential Long-Term Cost Savings

Recyclable materials can reduce disposal costs at end of product life and may offer supply chain resilience as recycled plastics become more widely available and competitively priced. If the plastic alternative can be sourced and processed efficiently, it may reduce Vent Ltd's exposure to metal commodity price volatility.

Disadvantage 1: Higher Unit Material Cost

The plastic alternative costs €0.35 per metre compared to the current €0.20 per metre standard – a 75% increase in material cost per metre. This would raise the standard material cost per unit of Part S from €1.00 (5m × €0.20) to €1.75 (5m × €0.35), directly reducing standard gross profit per unit and requiring management to decide whether to absorb the cost increase or pass it on through higher selling prices.

Disadvantage 2: Unknown Performance and Quality Risk

Part S is used in wind turbine construction, where structural reliability and safety are critical. Replacing a proven metal component with a plastic alternative introduces uncertainty around performance under mechanical stress, temperature variation, and long-term durability in outdoor environments. Any quality failures in the field could expose Vent Ltd to significant warranty claims, reputational damage, and potential liability.

Mark Allocation

Marks	Criteria
7–8	Two clearly explained advantages and two clearly explained disadvantages, each applied to Vent Ltd's specific context; cost impact of the price increase correctly quantified in at least one advantage/disadvantage.
5–6	All four points identified with reasonable explanation but largely generic rather than applied to Vent Ltd or the wind turbine sector specifically.
3–4	Three of four points adequately explained, or all four identified with minimal explanation.
1–2	One or two points identified with limited explanation.
0	No meaningful attempt.

Question 1(c)

10 marks

Vent Ltd uses a standard costing system to control costs, whereas a budgetary control system is often used as an alternative. Explain why a budgetary control system is preferred in certain business environments and illustrate your answer with an example.

Full Model Answer

Nature of Standard Costing vs Budgetary Control

Standard costing is most effective where a business produces a repetitive, homogeneous product or service under stable conditions – as is the case at Vent Ltd, where Part S is manufactured to a consistent specification. It sets a per-unit standard cost and uses variance analysis to identify deviations. Budgetary control, by contrast, operates at the level of the whole department, division, or organisation, setting overall spending limits and revenue targets rather than per-unit cost standards.

When Budgetary Control Is Preferred: Non-Repetitive or Service Environments

Budgetary control is preferred where output is not standardised and cannot easily be expressed as a uniform cost per unit. Professional service firms (e.g. law firms, consultancies, hospitals) deliver highly varied work where setting a meaningful "standard cost" per engagement or patient is impractical. In these environments, departmental budgets for salaries, materials, and overheads are set at the start of the period, and actual spending is compared to the total budget allowance rather than to a per-unit standard.

When Budgetary Control Is Preferred: Rapidly Changing Cost Environments

In industries where input prices, technology, or product mix change rapidly, standard costs become outdated quickly. A technology company launching new product variants every quarter, for instance, cannot rely on fixed standard costs that were set months earlier. Budgetary control allows management to plan and monitor costs flexibly on a period-by-period basis without the administrative burden of continuously revising standard cost cards.

Illustrative Example

A hospital provides an example: patient treatment times, drug costs, and staffing vary enormously across procedures. Setting a "standard cost per patient" would be meaningless given this complexity. Instead, the hospital sets departmental budgets (e.g. a total annual budget for the orthopaedic ward) and monitors actual expenditure against that budget monthly, identifying overall over- or under-spends by category rather than per-unit variances.

Mark Allocation

Marks	Criteria
9–10	Clear explanation of the distinction between standard costing and budgetary control; two or more well-explained business environments where budgetary control is preferred (non-repetitive output, rapidly changing costs, service industries); concrete, relevant example used to illustrate the point.
7–8	Correct explanation of why budgetary control is preferred in at least one environment, with a reasonable example; some comparison to standard costing present.
4–6	Reasonable understanding of budgetary control demonstrated but with limited explanation of when/why it is preferred over standard costing; example vague or missing.
1–3	Partial or generic description of budgetary control without adequate contrast with standard costing or identification of specific environments where it is preferred.
0	No meaningful attempt.

Question 1(d)

7 marks

Standard costs assist Vent Ltd in achieving financial objectives. Describe THREE non-financial objectives that the management of Vent Ltd could set, identifying the stakeholder group associated with each of the non-financial objectives you describe.

Full Model Answer

1. Product Quality and Defect Reduction — Stakeholder: Customers

Vent Ltd could set a non-financial objective to achieve a defect rate of less than 1% for Part S, measured by units returned or failed quality inspection. Wind turbine manufacturers (Vent Ltd's customers) require components that perform reliably under demanding conditions – quality failure could halt turbine installations and damage customer relationships. Measuring defect rates and setting reduction targets drives continuous improvement that supports customer retention without being purely financial in nature.

2. Employee Training and Development — Stakeholder: Employees

Vent Ltd could set an objective to ensure all production employees complete a minimum number of hours of technical skills training per year (e.g. 20 hours annually). Investing in workforce capability reduces labour inefficiency (as highlighted by the adverse labour efficiency variance in part (a)), improves morale and retention, and supports long-term productivity improvement. This objective directly benefits employees and is measured non-financially by training hours completed.

3. Environmental Sustainability — Stakeholder: Community / Society

Given Vent Ltd operates in the wind energy supply chain, it could set an objective to reduce the carbon footprint of its manufacturing process by a defined percentage per annum (e.g. 10% reduction in CO&sub2; emissions per unit produced). This is non-financial in nature, measured in tonnes of CO&sub2; or energy consumption per unit, and targets the wider community and society as stakeholders who are affected by industrial environmental impact.

Mark Allocation

Marks	Criteria
6–7	Three distinct non-financial objectives clearly described and applied to Vent Ltd's context; correct stakeholder group identified for each; objectives are genuinely non-financial (not reworded financial targets).
4–5	Three objectives identified with reasonable descriptions; stakeholders identified for at least two; some generic content but majority applied to Vent Ltd.
2–3	Two objectives adequately described with stakeholders identified, or three objectives listed with minimal explanation.
1	Only one objective properly described, or objectives are essentially financial in nature.
0	No meaningful attempt.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2

Konrad (Joe & Tom)

Total: 50 marks

This question tests investment appraisal for a partnership tendering to supply food huts and glamping pods at a 3-year outdoor festival. It covers NPV, ARR, Payback Period, an investment recommendation, cost classification (fixed/variable/mixed), and ESG goal-setting.

Parts in this Question

Part	Requirement	Marks
(a)	Calculate Net Present Value (NPV)	15
(b)	Calculate Accounting Rate of Return (ARR)	5
(c)	Calculate undiscounted Payback Period	5
(d)	Advise partners whether to proceed, using results from (a)–(c)	10
(e)	Classify marketing material costs, catering lease costs, tax advisory costs as fixed/variable/mixed	6
(f)	THREE ESG goals the partners should commit to, with practices to achieve each	9

Question 2(a)

15 marks

Calculate the Net Present Value (NPV) of the proposed festival project.

Full Model Answer

Key assumptions: The contract covers 3 festival seasons (2024, 2025, 2026 = Years 1–3). Discount factors are provided only for Years 1–3, confirming the NPV is computed over this 3-year contract period. Capital is paid on 1 January 2024 (Year 0). No marketing services income is included – the question asks Joe to provide a separate quote for that; it is a separate stream not part of this investment appraisal. Units used: 30 food huts and 80 glamping pods (Year 1 figures as stated; the question notes the "potential to increase" in later years but gives no specific numbers, so we use the stated quantities throughout).

Step 1: Year 0 capital outlay

Food huts: 30 × €1,900	€57,000
Glamping pods: 80 × €5,700	€456,000
Total capital outlay (Year 0)	**€513,000**

Step 2: Annual revenue (festival season = 3 months each year)

Food huts: 30 × €500/month × 3 months	€45,000
Glamping pods: 80 × €1,000/month × 3 months	€240,000
Total annual revenue	**€285,000**

Step 3: Annual operating costs

End of season deep clean – huts: 30 × €150	€4,500
End of season deep clean – pods: 80 × €200	€16,000
Off-season storage – huts: 30 × €90 × 9 months	€24,300
Off-season storage – pods: 80 × €170 × 9 months	€122,400
Total annual operating costs	**€167,200**

Step 4: Net annual cash flow and NPV calculation

	Annual CF (€)	Discount Factor	Present Value (€)
Year 1	117,800	0.926	109,082.80
Year 2	117,800	0.857	100,954.60
Year 3	117,800	0.794	93,533.20
Total PV of cash inflows			**303,570.60**
Less: Year 0 capital outlay			(513,000.00)
NET PRESENT VALUE			**€(209,429.40)**

Decision: The NPV is significantly negative at €(209,429.40). On a purely financial NPV basis the project does not generate sufficient discounted cash flows to recover the €513,000 capital investment at an 8% cost of capital, and would be **rejected**. See part (d) for a fuller recommendation to the partners.

Mark Allocation

Marks	Criteria
13–15	Correct capital outlay; correct annual revenue, all operating costs (cleaning + storage for both huts and pods) correctly computed; net annual CF of €117,800 correctly derived; discounting correctly applied using given factors; correct NPV of €(209,429.40) with clear accept/reject note.
10–12	Correct method throughout with one or two omitted cost items or minor arithmetic errors; structure and discounting correct; own-figure rule applied.
6–9	Correct NPV structure (Year 0 outlay vs discounted annual inflows) but with errors in multiple cost/revenue components; discounting attempted.
2–5	Some relevant cash flows identified and some discounting attempted, but significant errors in cash flow build-up or discount factor application.
0–1	No meaningful attempt.

Question 2(b)

5 marks

Calculate the Accounting Rate of Return (ARR) of the proposed festival project.

Full Model Answer

Annual net cash flow	€117,800
Annual depreciation (capital €513,000 ÷ 4 year useful life)	€128,250
Average annual accounting profit	€(10,450)
Initial investment	€513,000
ARR = (10,450) ÷ 513,000 × 100	**−2.04%**

Note on depreciation basis: The useful life is stated as 4 years with nil resale value, so straight-line depreciation = €513,000 ÷ 4 = €128,250 per annum. Since annual cash flow (€117,800) is less than annual depreciation (€128,250), the project generates an accounting loss on average – producing a negative ARR of −2.04%. Where a candidate uses average investment (€513,000 ÷ 2) as the denominator, full own-figure credit is awarded provided the method is clearly shown.

Mark Allocation

Marks	Criteria
5	Correct depreciation calculated using 4-year life and nil residual; correct average annual profit (or loss); correct ARR of −2.04% (or correct own-figure from part (a) net CF).
3–4	Correct method applied with a minor error (e.g. 3-year depreciation used instead of 4, or average investment basis used – both acceptable with correct method shown).
2	Correct formula identified but inputs incorrectly derived (e.g. using gross revenue rather than net profit).
1	Some understanding of ARR shown but substantially misapplied.
0	No meaningful attempt.

Question 2(c)

5 marks

Calculate the undiscounted Payback Period of the proposed festival project.

Full Model Answer

Year	Net Cash Flow (€)	Cumulative (€)
0	(513,000)	(513,000)
1	117,800	(395,200)
2	117,800	(277,400)
3	117,800	(159,600)
4	117,800	(41,800)
5	117,800	76,000

Payback occurs during Year 5. At the start of Year 5 there is €41,800 still to recover.

Amount unrecovered at start of Year 5	€41,800
Year 5 net cash flow	€117,800
Fraction of Year 5 required	41,800 ÷ 117,800 = 0.355
Payback Period	**4 years + 0.355 years ≈ 4.35 years (approx 4 years 4 months)**

Critical note: The contract runs for only 3 years. Payback is not achieved within the contract period (cumulative cash flow is still €(159,600) at the end of Year 3). This is a significant financial concern raised in the recommendation in part (d).

Mark Allocation

Marks	Criteria
5	Correct cumulative cash flow table; correct identification of payback falling in Year 5; correct fractional-year calculation; correct payback period of approx 4.35 years; note that this exceeds the 3-year contract.
3–4	Correct method and correct year of payback; minor arithmetic error in fractional year, or note about 3-year contract missing.
2	Correct cumulative approach but year of payback incorrectly identified, or fractional year calculation omitted.
1	Attempt made but cumulative cash flows substantially wrong.
0	No meaningful attempt.

Question 2(d)

10 marks

Using the results from your calculations in (a) to (c) above, advise the Konrad partners whether to proceed with tendering for the project, explaining your answer in detail.

Full Model Answer

Summary of Financial Results

All three investment appraisal methods point against proceeding with the project as currently structured. The NPV is deeply negative at €(209,429.40), indicating the project destroys value at an 8% cost of capital. The ARR is −2.04%, meaning the project on average generates an accounting loss rather than a profit. The payback period of approximately 4.35 years significantly exceeds the 3-year festival contract, meaning the initial capital outlay will not be recovered within the life of the contract.

Specific Financial Concern: Off-Season Storage Costs

The principal financial problem is the very high off-season storage costs: €146,700 per annum (huts €24,300 + pods €122,400), which consume a large portion of the €285,000 annual rental revenue. The partners should explore whether storage costs could be reduced – for example, by negotiating a lower storage rate, finding alternative storage, or structuring the contract so the festival promoter provides storage.

Consideration: Revenue Upside in Years 2 and 3

The question notes the festival promoters estimate that the number of huts and pods needed could increase in Years 2 and 3 depending on festival success. If, for example, huts and pods doubled in Year 2 and 3, revenues and cash flows would scale up significantly, potentially improving NPV, ARR, and payback. The partners should model this scenario carefully before deciding. However, this upside is uncertain and should not be the primary basis for a recommendation.

Recommendation

On the basis of the financial analysis, the partners should **not proceed** with tendering as currently structured. The negative NPV and ARR, combined with a payback period exceeding the contract term, indicate the project is financially unviable at current cost and revenue levels. Before rejecting the opportunity entirely, the partners should renegotiate off-season storage arrangements to reduce this major cost driver, and model the upside scenario of increased festival scale in Years 2 and 3 to establish whether the project could become viable under revised assumptions.

Mark Allocation

Marks	Criteria
9–10	Clear reference to all three appraisal results from (a)–(c) with correct interpretation; specific financial concern (storage costs, payback exceeding contract) identified; clear, justified recommendation; consideration of upside potential acknowledged.
7–8	Two or three appraisal results correctly interpreted and used in the recommendation; clear reject recommendation with financial justification; one or more upside/contextual factors considered.
4–6	Recommendation given but only loosely linked to the financial results; limited analysis of the specific financial drivers (e.g. storage costs) or the contract period vs payback issue.
1–3	A recommendation is stated but with minimal reference to the quantitative results from parts (a)–(c).
0	No meaningful attempt.

Question 2(e)

6 marks

Advise the partners whether each of the following costs would be classified as a fixed cost, variable cost or overlapping/mixed cost. For each cost explain your answer fully: Marketing material costs paid by Joe; Catering equipment lease costs; Tax advisory costs.

Full Model Answer

Marketing Material Costs — Variable Cost

Marketing material costs (e.g. brochures, promotional materials, social media spend) are likely to vary with the scale of the festival operation. As the number of festivals, seasons, or huts/pods increases, the volume of marketing materials required would increase proportionally. These costs have no irreducible fixed minimum – if the partnership did not operate in a given year, marketing spend could be zero. Therefore, marketing material costs are classified as **variable**.

Catering Equipment Lease Costs — Fixed Cost (or Mixed)

A lease contract typically specifies a fixed periodic payment (e.g. monthly or annual) regardless of how many units of catering equipment are actually used or how many customers are served. The lease payment does not change with the level of activity within the contract term, making it a **fixed cost**. However, if the lease includes a usage-based component (e.g. a base monthly charge plus a per-event surcharge), it would be classified as a **mixed cost** – with a fixed element (the base lease) and a variable element (the usage charge). Without further information, a fixed cost classification is most appropriate.

Tax Advisory Costs — Fixed Cost

Tax advisory fees (e.g. annual tax returns, compliance advice) are typically charged on a fixed retainer or agreed flat fee basis, regardless of the volume of business undertaken by the partnership in the year. The cost does not vary with the number of huts, pods, or customers served. Therefore, tax advisory costs are classified as a **fixed cost**.

Mark Allocation

Marks	Criteria
5–6	All three costs correctly classified with a clear, well-reasoned explanation for each; recognition that catering lease could be fixed or mixed depending on contract terms, with justification.
3–4	Two or three costs correctly classified but with limited explanation, or the mixed-cost nuance for the catering lease not recognised.
2	Two costs correctly classified with brief justification.
1	One cost correctly classified, or all three classified without meaningful explanation.
0	No meaningful attempt.

Question 2(f)

9 marks

The festival promotor has a strong Environmental, Social and Governance (ESG) policy and a reputation in the entertainment industry for being environmentally conscious. Suggest THREE goals that, assuming they proceed with the tender, the partners should commit to achieving to be in line with the festival promotion company's ESG policy. Include in your answer practices needed to achieve these goals.

Full Model Answer

1. Environmental: Zero Single-Use Plastic in Food Hut Operations

Goal: Eliminate all single-use plastics from the food hut catering operations by Year 1 of the contract. Practices: Source all packaging from certified compostable or reusable materials; brief all catering staff on single-use plastic policy before the festival season opens; work with suppliers to ensure delivery packaging is recyclable; provide clearly labelled recycling and composting stations at each food hut area; measure and report on waste volumes per season.

2. Social: Local Employment and Community Benefit

Goal: Source a minimum of 60% of seasonal staff from the local community surrounding the festival site. Practices: Advertise seasonal positions through local job centres, schools, and community boards rather than exclusively through national platforms; partner with local training providers to offer food safety and hospitality skills development; pay at or above the living wage to all seasonal workers; donate a percentage of seasonal profits to a local environmental charity each year.

3. Governance: Transparent ESG Reporting and Supply Chain Ethics

Goal: Publish an annual ESG report covering the Konrad partnership's environmental impact, employment practices, and governance standards for each festival season. Practices: Maintain records of energy consumption, waste volumes, and supplier ethics throughout the season; conduct supplier due diligence to ensure food and materials suppliers meet fair trade or equivalent standards; present the ESG report to the festival promoter at the end of each season as part of the contract review process.

Mark Allocation

Marks	Criteria
8–9	Three distinct ESG goals clearly spanning Environmental, Social, and Governance dimensions; each goal is specific, measurable, and applied to the festival/catering context; concrete practices included for each goal.
6–7	Three goals identified covering at least two of the three ESG dimensions; goals are relevant to the context; practices included for each but less detailed.
4–5	Two well-developed goals with practices, or three goals identified with minimal practices described.
1–3	One or two goals identified; practices vague or absent; goals are generic rather than applied to the festival/food hut context.
0	No meaningful attempt.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3

JXC Ltd

Total: 50 marks

This question tests production overhead allocation – first on a single company-wide rate (direct labour hours), then departmental rates using the most appropriate basis for each department. It also covers commentary on the two approaches, managing customer returns, value-added analysis, and the Balanced Scorecard framework.

Parts in this Question

Part	Requirement	Marks
(a)	Calculate company-wide production overhead allocation rate (existing DLH basis)	8
(b)	Calculate departmental rates for Assembly, Wiring and Finishing on the most appropriate basis	15
(c)	Comment on results from (a) and (b) and implications for 2023 production planning	7
(d)	Steps to address increase in returned goods by customers	5
(e)	Assess viability of colour customisation option using value-added/non-value-added framework	5
(f)	Balanced Scorecard: TWO objectives and measures for each of the three non-financial perspectives	10

Question 3(a)

8 marks

Calculate JXC 2022 production overhead allocation rate based on the existing policy of allocating overheads.

Full Model Answer

The existing policy allocates overheads based on **direct labour hours**. The company-wide rate is calculated by dividing total overheads across all three departments by total direct labour hours across all three departments.

Department	Overhead (€)	Machine Hours	Direct Labour Hours
Assembly	31,800	5,500	1,300
Wiring	25,640	750	2,660
Finishing	28,500	1,780	950
Total	**85,940**	**8,030**	**4,910**

Total production overheads	€85,940
Total direct labour hours	4,910 hours
Company-wide overhead rate	**€85,940 ÷ 4,910 = €17.50 per direct labour hour**

Mark Allocation

Marks	Criteria
7–8	Correct identification of DLH as the existing basis; correct totalling of all three departments’ overheads (€85,940) and DLH (4,910); correct company-wide rate of €17.50 per DLH with workings shown.
5–6	Correct method but minor arithmetic error in totalling, or rate calculated to a slightly different decimal; DLH basis correctly identified.
3–4	Correct formula applied (total OH ÷ total activity) but wrong activity basis used (e.g. machine hours instead of DLH).
1–2	Partial attempt; some relevant figures identified but rate calculation substantially wrong.
0	No meaningful attempt.

Question 3(b)

15 marks

Calculate JXC 2022 departmental production overhead allocation rates for the THREE production departments based on the most appropriate basis for each department.

Full Model Answer

The most appropriate basis is selected by comparing machine hours to direct labour hours in each department – where machine hours dominate, a **machine hour rate** is more accurate; where labour hours dominate, a **direct labour hour rate** is more appropriate.

Department	Machine Hrs	DLH	Dominant Activity	Appropriate Basis
Assembly	5,500	1,300	Machine-intensive (5,500 >> 1,300)	Machine hour rate
Wiring	750	2,660	Labour-intensive (2,660 >> 750)	Direct labour hour rate
Finishing	1,780	950	Machine hours exceed DLH	Machine hour rate

Departmental rates

Department	Basis	Overhead (€)	Hours	Rate
Assembly	Machine hours	31,800	5,500	€5.78 per machine hour
Wiring	Direct labour hours	25,640	2,660	€9.64 per DLH
Finishing	Machine hours	28,500	1,780	€16.01 per machine hour

Note on Finishing department: Finishing includes quality inspection and full testing, which are labour-intensive activities. A candidate who selects DLH as the basis for Finishing on these grounds should be awarded full marks provided the reasoning is clearly explained – this is an accepted alternative interpretation. The DLH rate for Finishing would be: €28,500 ÷ 950 = €30.00 per DLH.

Mark Allocation

Marks	Criteria
13–15	All three departments' most appropriate bases correctly justified with reference to the relative size of machine hours vs DLH; correct rates calculated for all three (Assembly €5.78/MH, Wiring €9.64/DLH, Finishing €16.01/MH or €30.00/DLH with justification); workings shown.
10–12	Two of three departments correctly justified and rate calculated; third department either incorrect basis or minor arithmetic error; own-figure rule applied.
6–9	At least one department correctly justified and rate calculated; the concept of selecting the appropriate basis (comparing machine vs DLH activity) demonstrated even if applied incorrectly in places.
2–5	Rates calculated but all on the same basis (e.g. all DLH or all machine hours) without differentiation; some understanding of the mechanics shown.
0–1	No meaningful attempt.

Question 3(c)

7 marks

Comment on the results obtained in (a) and (b) and suggest how the JXC production manager could use this information in planning for 2023 production.

Full Model Answer

Distortion Under the Company-Wide Rate

The single company-wide rate of €17.50 per DLH significantly distorts product costs by treating all overhead as if it were driven by direct labour hours. Assembly, the most machine-intensive department (5,500 machine hours vs 1,300 DLH), is severely *under-charged* under the company-wide DLH rate – only €22,754 is absorbed against an actual overhead of €31,800. By contrast, Wiring (labour-intensive) is heavily *over-charged* – absorbing €46,558 against actual overhead of only €25,640. This cross-subsidisation means any speaker that passes through Assembly intensively will have its cost understated, while labour-heavy wiring work will be overstated.

Implications for 2023 Production Planning

Using departmental rates in 2023 will give the production manager a far more accurate picture of the true cost of each speaker model, particularly models that spend disproportionate time in the high-overhead Finishing department (€16.01/machine hour) compared to Assembly (€5.78/machine hour). This improved cost accuracy supports better pricing decisions, helps identify which product variants are genuinely profitable after accurate overhead allocation, and allows the manager to monitor overhead recovery more precisely at department level – spotting where overheads are increasing relative to budgeted activity before year-end.

Mark Allocation

Marks	Criteria
6–7	Clear explanation of the distortion caused by the company-wide rate with specific reference to the numerical results (e.g. Assembly under-absorbed, Wiring over-absorbed); practical implications for 2023 planning (pricing, cost control, overhead monitoring) clearly articulated.
4–5	Comment on distortion between the two approaches with reasonable explanation; planning implication identified but less developed.
2–3	General acknowledgement that departmental rates are more accurate without specific reference to the numerical results or specific planning implications.
1	Vague reference to departmental rates being “better” without explanation.
0	No meaningful attempt.

Question 3(d)

5 marks

Suggest steps that the management of JXC could take to address the increase in returned goods by customers.

Full Model Answer

Investigate Root Causes Through Returns Data Analysis

JXC should first systematically analyse the reasons for returns – tracking whether returns are driven by defects, incorrect orders, cosmetic issues (e.g. colour customisation errors), or performance failures. A returns log coding system that categorises each return by cause will identify whether the increase is concentrated in a particular product variant, production batch, or time period, enabling targeted corrective action rather than broad, unfocused responses.

Strengthen Finishing Department Quality Inspection Process

The Finishing department already includes quality inspection and full testing before items move to packing. JXC should review whether the inspection criteria are sufficiently rigorous and consistently applied – particularly for the solar charging component and the colour customisation process (which is undertaken by 40% of customers and may be a source of specification mismatch or quality variation). Introducing a second-check process or statistical sampling at packing could catch defects before despatch.

Supplier and Materials Quality Review

If returns relate to component failures (e.g. the solar charger), JXC should audit supplier quality standards, review incoming goods inspection procedures, and consider whether any supply chain changes made in 2022 correlate with the timing of the returns increase. Switching to or qualifying a secondary supplier for critical components could reduce quality risk.

Mark Allocation

Marks	Criteria
5	Three or more distinct, practical steps identified and explained, applied to JXC's specific context (speaker with solar charger, colour customisation, Finishing department quality inspection); logical progression from diagnosis to corrective action.
3–4	Two well-explained steps applied to JXC's context, or three steps identified with less detail.
2	Two steps identified with brief explanation; partially applied to JXC's context.
1	One step identified, or steps are entirely generic with no application to JXC.
0	No meaningful attempt.

Question 3(e)

5 marks

Suggest to the JXC finance team how it might assess the viability of continuing to offer the colour customisation option in the context of Value Added and Non-Value Added activities.

Full Model Answer

Identify Value-Added vs Non-Value-Added Elements of the Customisation Process

A **value-added activity** is one that the customer is willing to pay for and that directly contributes to meeting customer requirements. The colour customisation itself – the actual application of the chosen colour – is value-added, since 40% of customers pay €150 specifically for this option. **Non-value-added activities** are those that consume resources but do not add value in the customer's eyes: for example, rework caused by colour application errors, waiting time between the main production run and the customisation step, or re-inspection after customisation is complete.

Financial Viability Assessment

The finance team should calculate the full incremental cost of the colour customisation process – including direct material cost for colour, additional labour time in Finishing, any specialist equipment depreciation, the cost of rework/returns caused by colour errors, and any quality inspection time directly attributable to the option. If the total incremental cost per unit is less than €150, the option adds financial value and should be continued. If non-value-added activities (rework, re-inspection, waiting) are found to be driving the incremental cost above €150, these sub-activities should be targeted for elimination before discontinuing the option entirely.

Mark Allocation

Marks	Criteria
5	Clear definition of value-added vs non-value-added activities applied to the customisation option; identification of both value-added (the customisation itself, revenue-generating) and non-value-added (rework, waiting, re-inspection) elements; practical financial viability assessment linked to the €150 incremental revenue.
3–4	Correct distinction between VA and NVA with reasonable application to JXC; financial viability mentioned but less fully developed.
2	Correct definitions of VA and NVA given but applied generically rather than to the colour customisation specifically.
1	Partial understanding of VA/NVA concepts demonstrated without application to JXC.
0	No meaningful attempt.

Question 3(f)

10 marks

The management of JXC is considering adopting the Balanced Scorecard framework. Describe TWO objectives and the corresponding measures that you consider the most important for JXC to focus on for each of the three non-financial perspectives.

Full Model Answer

The three non-financial Balanced Scorecard perspectives are: **Customer**, **Internal Business Process**, and **Learning & Growth**. (Note: the Financial perspective is the fourth BSC perspective but is not a "non-financial" perspective and is excluded from this question.)

Customer Perspective

Objective 1: Reduce customer returns and improve product quality perception. Measure: Percentage of units returned per quarter; customer satisfaction score from post-purchase survey. Given JXC's noted increase in returns in 2022, this is a critical customer-facing metric that directly impacts brand reputation in the premium speaker market.

Objective 2: Grow customer base through the colour customisation offering. Measure: Percentage of new customers who selected the colour customisation option; repeat purchase rate among customisation customers. Since 40% of customers already choose customisation, tracking adoption and retention of this premium segment supports revenue growth strategy.

Internal Business Process Perspective

Objective 1: Improve production efficiency and reduce defect rate. Measure: Number of defects identified at Finishing quality inspection per 100 units produced; percentage of units passing quality inspection first time (first-pass yield). This directly addresses the quality concern raised in parts (d) and (e).

Objective 2: Reduce production overhead cost per unit through better resource utilisation. Measure: Actual overhead absorbed vs budgeted overhead per department (using the departmental rates from part (b)); machine utilisation rate in Assembly. This supports the shift to departmental overhead rates and drives more accountable cost management.

Learning & Growth Perspective

Objective 1: Develop technical skills for solar charger integration and new product innovation. Measure: Number of engineering training hours per employee per year; number of new product features developed or tested per year. As a high-end electronics producer competing on innovation, continuous technical capability development is critical.

Objective 2: Improve employee engagement and retention. Measure: Annual employee satisfaction score; staff turnover rate. High turnover in specialist assembly or finishing roles directly impacts product quality consistency and training costs.

Mark Allocation

Marks	Criteria
9–10	Two objectives and corresponding measures correctly provided for all three non-financial perspectives (Customer, Internal Process, Learning & Growth); objectives are specific, realistic, and applied to JXC’s context (speaker manufacturing, returns issue, customisation); measures are quantifiable and linked to each objective.
7–8	Two objectives and measures for all three perspectives, but one or two are generic rather than applied to JXC, or one measure is vague/not clearly quantifiable.
4–6	All three perspectives addressed but only one objective per perspective adequately developed, or two perspectives well covered and one underdeveloped.
2–3	Only two of the three non-financial perspectives addressed, or all three addressed with very generic content and no application to JXC.
0–1	No meaningful attempt, or financial perspective used instead of non-financial perspectives.

QUESTION 3 TOTAL — 50 MARKS